# Hospital Patient Records ETL Pipeline

MySQL (OLTP) → Pentaho PDI → PostgreSQL (OLAP)

Project Documentation | github.com/priya-2527

| 100,000 | 89,386 | 25,328 | 6 | Daily |
|---|---|---|---|---|
| Raw Records | Clean Records | Nulls Cleaned | New Columns | Automated |

## 1. Project Overview

End-to-end ETL pipeline built with Pentaho PDI. Extracts 100,000 raw hospital patient records from MySQL (OLTP), applies data cleaning and transformation, loads clean data into PostgreSQL (OLAP). Fully automated via Pentaho Job + Windows Task Scheduler.

| Component | Technology | Role |
|---|---|---|
| Source DB | MySQL 8.0 | OLTP — Raw patient data |
| ETL Tool | Pentaho PDI 9.4 | Extract, Transform, Load |
| Destination | PostgreSQL 18 | OLAP — Clean data warehouse |
| Scheduler | Windows Task Scheduler | Daily automation |
| Runner | Kitchen.bat | Command line executor |

## 2. Pipeline Architecture

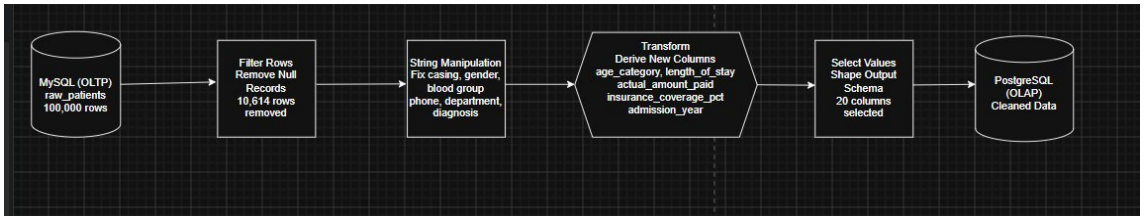



*Figure 1: ETL pipeline architecture diagram (Draw.io)*

## 3. Source Data

100,000 raw records with mixed casing, inconsistent values, abbreviations, special characters and nulls.

### Row Count

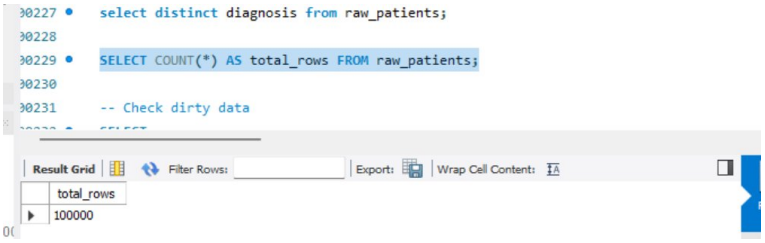
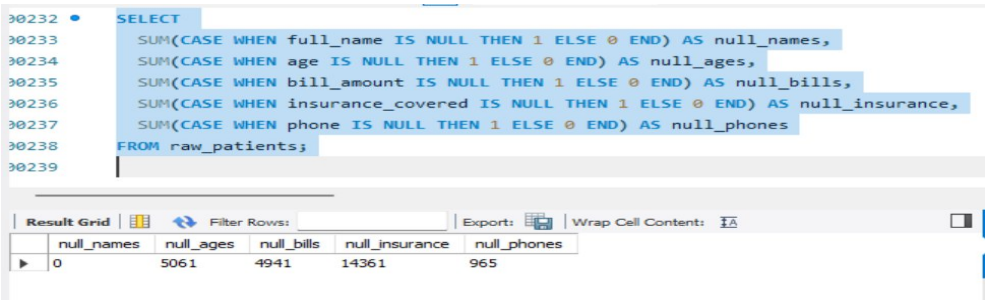



*Figure 2: 100,000 records confirmed in MySQL*

### Null Analysis



```
SELECT
  SUM(CASE WHEN full_name IS NULL THEN 1 ELSE 0 END) AS null_names,
  SUM(CASE WHEN age IS NULL THEN 1 ELSE 0 END) AS null_ages,
  SUM(CASE WHEN bill_amount IS NULL THEN 1 ELSE 0 END) AS null_bills,
  SUM(CASE WHEN insurance_covered IS NULL THEN 1 ELSE 0 END) AS null_insurance,
  SUM(CASE WHEN phone IS NULL THEN 1 ELSE 0 END) AS null_phones
FROM raw_patients;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

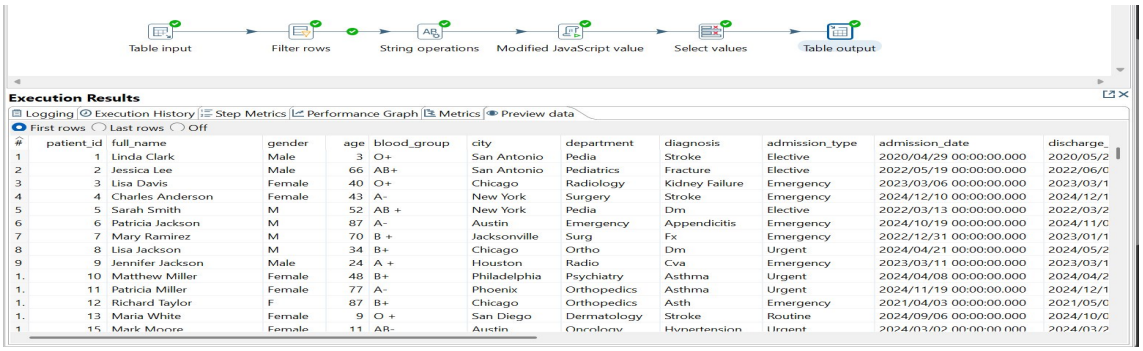
| null_names | null_ages | null_bills | null_insurance | null_phones |
|---|---|---|---|---|
| 0 | 5061 | 4941 | 14361 | 965 |

*Figure 3: 25,328 total nulls across 5 columns*

| Column | Nulls | Action |
|---|---|---|
| age | 5,061 | Filtered out |
| bill_amount | 4,941 | Filtered out |
| insurance_covered | 14,361 | actual_amount_paid = bill when null |
| phone | 965 | Filtered out |
| full_name | 0 | No action |

# 4. Pentaho Pipeline

6-step transformation built in Pentaho Spoon. All steps show green ticks after successful run.




| # | patient_id | full_name | gender | age | blood_group | city | department | diagnosis | admission_type | admission_date | discharge_ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Linda Clark | Male | 3 | O+ | San Antonio | Pedia | Stroke | Elective | 2020/04/29 00:00:00.000 | 2020/05/2 |
| 2 | 2 | Jessica Lee | Male | 66 | AB+ | San Antonio | Pediatrics | Fracture | Elective | 2022/05/19 00:00:00.000 | 2022/06/0 |
| 3 | 3 | Lisa Davis | Female | 40 | O+ | Chicago | Radiology | Kidney Failure | Emergency | 2023/03/06 00:00:00.000 | 2023/03/1 |
| 4 | 4 | Charles Anderson | Female | 43 | A- | New York | Surgery | Stroke | Emergency | 2024/12/10 00:00:00.000 | 2024/12/1 |
| 5 | 5 | Sarah Smith | M | 52 | AB + | New York | Pedia | Dm | Elective | 2022/03/13 00:00:00.000 | 2022/03/2 |
| 6 | 6 | Patricia Jackson | M | 87 | A- | Austin | Emergency | Appendicitis | Emergency | 2024/10/19 00:00:00.000 | 2024/11/0 |
| 7 | 7 | Mary Ramirez | M | 70 | B + | Jacksonville | Surg | Fx | Emergency | 2022/12/31 00:00:00.000 | 2023/01/1 |
| 8 | 8 | Lisa Jackson | M | 34 | B+ | Chicago | Ortho | Dm | Urgent | 2024/04/21 00:00:00.000 | 2024/05/2 |
| 9 | 9 | Jennifer Jackson | Male | 24 | A + | Houston | Radio | Cva | Emergency | 2023/03/11 00:00:00.000 | 2023/03/1 |
| 1. | 10 | Matthew Miller | Female | 48 | B+ | Philadelphia | Psychiatry | Asthma | Urgent | 2024/04/08 00:00:00.000 | 2024/04/2 |
| 1. | 11 | Patricia Miller | Female | 77 | A- | Phoenix | Orthopedics | Asthma | Urgent | 2024/11/19 00:00:00.000 | 2024/12/1 |
| 1. | 12 | Richard Taylor | F | 87 | B+ | Chicago | Orthopedics | Asth | Emergency | 2021/04/03 00:00:00.000 | 2021/05/0 |
| 1. | 13 | Maria White | Female | 9 | O + | San Diego | Dermatology | Stroke | Routine | 2024/09/06 00:00:00.000 | 2024/10/0 |
| 1 | 15 | Mark Moore | Female | 11 | AB- | Austin | Oncology | Hypertension | Urgent | 2024/03/02 00:00:00.000 | 2024/03/2 |

*Figure 4: Pentaho PDI canvas — 6 steps with green success ticks*

| | |
|---|---|
| 1 | **Table Input** — Extracts all 100,000 records from MySQL via SELECT * FROM raw_patients. |
| 2 | **Filter Rows** — Removes 10,614 null records from age, bill_amount and phone columns. |
| 3 | **String Operations** — Cleans 9 columns: trim, InitCap, uppercase blood groups, remove phone special chars. |
| 4 | **Modified JavaScript** — Derives 6 new columns: age_category, actual_amount_paid, length_of_stay, etc. |
| 5 | **Select Values** — Selects and renames 20 final columns for PostgreSQL schema. |
| 6 | **Table Output** — Loads 89,386 clean records into PostgreSQL clean_patients table. |

## 5. String Manipulation

| Column | Operation | Before | After |
|---|---|---|---|
| full_name | Trim + InitCap | JOHN smith | John Smith |
| gender | Trim + InitCap | MALE/male/M | Male |
| blood_group | Trim + Upper | a+, AB + | A+, AB+ |
| phone | Remove Special | (418) 566-7216 | 4185667216 |
| department | Trim + InitCap | cardio | Cardio |
| diagnosis | Trim + InitCap | HTN | Htn |
| doctor_name | Trim + InitCap | dr. JOHN smith | Dr. John Smith |
| city | Trim + InitCap | new york | New York |
| admission_type | Trim + InitCap | EMERGENCY | Emergency |

## 6. Derived Columns

| New Column | Logic | Sample |
|---|---|---|
| age_category | Child(<=12)/Teenager/Young Adult/Adult/Senior(61+) | Adult |
| actual_amount_paid | bill_amount - insurance_covered | 1776.76 |
| length_of_stay | discharge_date - admission_date (days) | 7 |
| insurance_coverage_pct | (insurance/bill)*100 | 49.0 |
| admission_year | Year from admission_date | 2022 |
| clean_phone | Remove (, ), -, spaces | 4185667216 |

## 7. Results

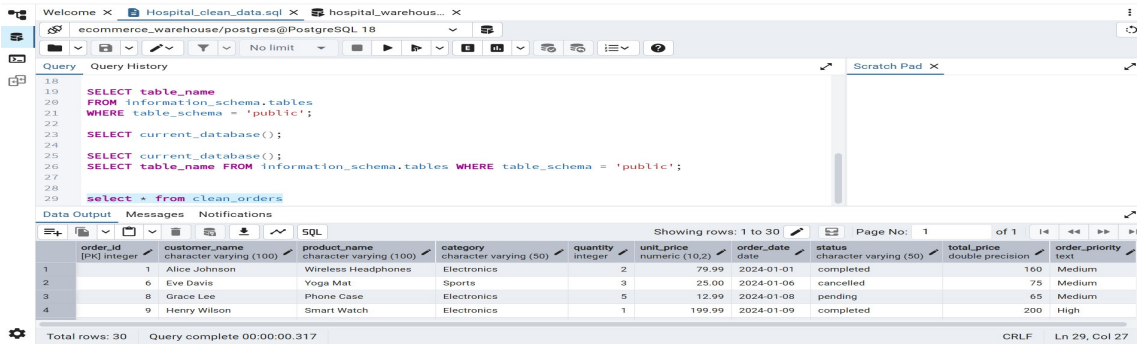



*Figure 5: Clean data in PostgreSQL clean_patients table*

| Metric | Value |
|---|---|
| Raw records extracted | **100,000** |
| Null records removed | **10,614** |
| Clean records in PostgreSQL | **89,386** |
| Null values cleaned | **25,328** |
| Columns string manipulated | **9** |
| New columns derived | **6** |
| Automation | **Daily — Zero manual intervention** |

## 8. Automation

Transformation wrapped in a Pentaho Job and scheduled via Kitchen.bat + Windows Task Scheduler to run daily automatically.

*Figure 6: Pentaho Job — Start → Transformation → Success*

| Setting | Value |
|---|---|
| Job Structure | Start → Transformation → Success |
| Job File | hospital_job.kjb |
| Transformation | hospital_pipeline.ktr |
| Runner | Kitchen.bat |
| Scheduler | Windows Task Scheduler |
| Frequency | Daily — Fully Automated |

## 9. Tools & Technologies

| Tool | Version | Purpose |
|---|---|---|
| Pentaho PDI | 9.4 | ETL pipeline |
| MySQL | 8.0 | OLTP source |
| PostgreSQL | 18 | OLAP destination |
| Kitchen.bat | Built-in | Job runner |
| Windows Task Scheduler | Win 11 | Automation |
| Draw.io | Web | Architecture diagram |